

Skills Summary

Adding and Subtracting Mixed Numbers

Use this reference while completing your worksheet or when studying.

A mixed number is a whole plus a fraction, so work the two parts in columns: wholes with wholes, fractions with fractions, then put them back together.

Same-size pieces rule: fractions can only be added or subtracted when the denominators match. The denominator names the size of the piece and never changes when you add or subtract.

One whole is always the fraction with equal top and bottom — three thirds, four fourths, eight eighths. That is what makes trading possible in both directions.

1. Adding with the same denominator

Add the wholes, add the fractions, and then look at the fraction part. If it is one whole or more, trade it: carry the whole across and keep what is left. Finish in simplest form, with the unit from the story.

Example: A hike is $3\frac{2}{5}$ km, then $1\frac{4}{5}$ km. How far in all?

Step 1. The fifths match, so the parts can be added as they stand — but two fifths plus four fifths passes one whole, so expect a trade.

Step 2. Wholes: $3 + 1 = 4$. Fifths: $\frac{2}{5} + \frac{4}{5} = \frac{6}{5} = 1\frac{1}{5}$, and $4 + 1\frac{1}{5}$ is the total.
The hike is $5\frac{1}{5}$ km.

Try it: A ribbon of $2\frac{3}{8}$ m is joined to one of $4\frac{6}{8}$ m. How long is the joined ribbon?

$\frac{8}{1}$ check

2. Subtracting when you have to regroup

If the fraction you are taking away is bigger than the one you have, borrow one whole and change it into a fraction with the same denominator, then add it to the fraction you already had.

Never turn the subtraction around. Reversing the fraction column answers a different question and always gives an answer that is too big.

Example: A jug holds $5\frac{1}{4}$ L. Pour out $1\frac{3}{4}$ L. How much is left?

Step 1. Three quarters cannot come out of one quarter, so borrow a whole before subtracting anything.

Step 2. $5\frac{1}{4} = 4\frac{5}{4}$. Wholes: $4 - 1 = 3$. Quarters: $\frac{5}{4} - \frac{3}{4} = \frac{2}{4} = \frac{1}{2}$.
There is $3\frac{1}{2}$ L left.

Try it: A board is $6\frac{1}{5}$ ft long. A piece of $2\frac{3}{5}$ ft is cut off. How much board is left?

$\frac{5}{3}$ check

3. When the denominators are different

Rewrite both fractions with a common denominator first — the smallest number both denominators divide is easiest — and only then add or subtract in columns.

After rewriting, check the fraction column again: a subtraction may still need regrouping, and a sum may still need a trade.

Example: A cook mixes $2\frac{1}{2}$ cups of stock with $1\frac{1}{3}$ cups of cream. How much liquid is that?

Step 1. Halves and thirds are different-sized pieces, so nothing can be added until both are written over one denominator, and sixths is the smallest that works.

Step 2. $\frac{1}{2} = \frac{3}{6}$ and $\frac{1}{3} = \frac{2}{6}$. Wholes: $2 + 1 = 3$. Sixths: $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.

The mixture is $3\frac{5}{6}$ cups.

Try it: A tank holds $4\frac{2}{3}$ gal and $1\frac{1}{4}$ gal is drained. How much is left?

Check: $\frac{21}{9}$ 8 :xəpə

(!) **Watch out:** adding or subtracting straight across the denominators is the most common slip. Estimate with the nearest whole numbers before you start, and check that your answer lands near the estimate.